

 Section: Particles and Waves

 Topic: Wave–Particle Duality – The Photoelectric Effect

Question 9 – Analogy for the Photoelectric Effect

Prompt from the paper:

“It’s like putting money into a vending machine.
You won’t get your snack unless you have enough money,
no matter how many coins you put in.
If you put in too much money, your snack will come out and you will get change back.”

 **3 marks – Physics-Based Commentary:**

 **Answer aligned with the marking instructions:**

This is a **good analogy** for the photoelectric effect:

- The **coins** represent **photons**.
- The **snack** represents **an electron being ejected**.
- The idea that you need **enough money in a single coin** reflects the requirement that a **photon must have energy greater than the work function** to eject an electron.
- Adding **more photons** (i.e. intensity) **does not cause emission** unless each one individually has **enough energy** — just like multiple small coins won’t release a snack if none are large enough.
- The idea of **getting change** after a large coin fits with **excess photon energy becoming the kinetic energy of the electron**.

 **Conclusion:**

The vending machine analogy effectively conveys the **threshold frequency** and the idea that photon energy must be above a certain minimum. It also captures how **surplus energy** becomes **kinetic energy** of the ejected electron.

 Revision Tips

- **Work function:** minimum energy needed to release an electron from a metal surface
- **Photon energy:**

$E = hf$

- **Photoelectric equation:**

$hf = \Phi + E_k$

where Φ is the work function and E_k is kinetic energy